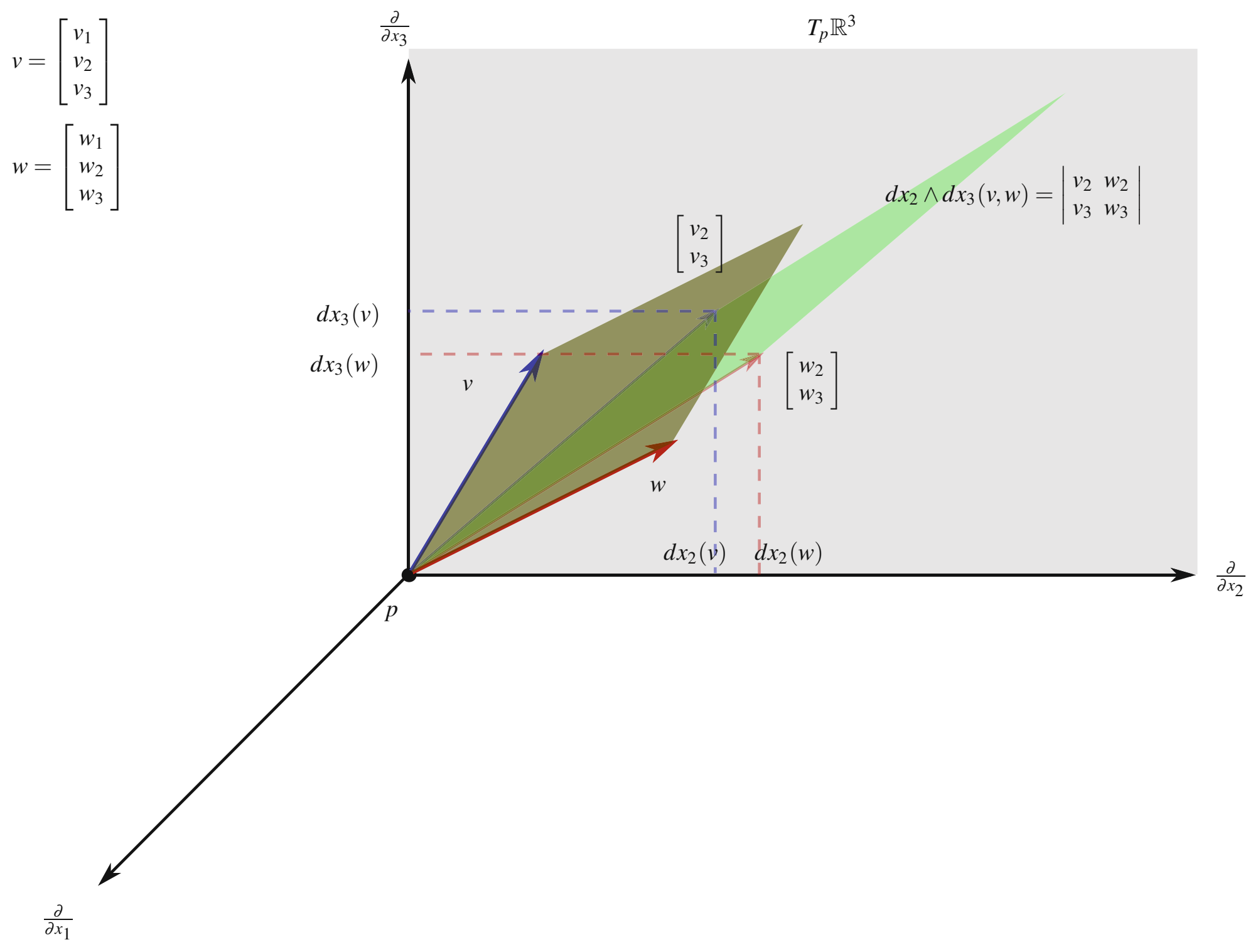

$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$
$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$
$\frac{\partial}{\partial x_3}$
$T_p\mathbb{R}^3$
$dx_2 \wedge dx_3(v,w) = \begin{vmatrix} v_2 & w_2 \\ v_3 & w_3 \end{vmatrix}$
$\begin{bmatrix} v_2 \\ v_3 \end{bmatrix}$
$dx_3(v)$
$dx_3(w)$
v
$\begin{bmatrix} w_2 \\ w_3 \end{bmatrix}$
w
$dx_2(v)$
$dx_2(w)$
$\frac{\partial}{\partial x_2}$
p
$\frac{\partial}{\partial x_1}$